

SmartBuild AI Technical Explanation

Data Model Predictions, Evaluation and Responsible Positioning

Purpose

This document explains what SmartBuild AI does, how its data is generated, which machine-learning techniques are used, how predictions and anomalies are produced, and how model performance should be interpreted. It also identifies the limits of the current prototype and provides accurate language for professional presentation.

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Executive Summary

SmartBuild AI is a working machine-learning prototype that estimates expected hourly electrical energy use in a commercial building. It then compares expected consumption with actual consumption and flags unusually large differences for investigation. The application combines a reproducible data generator, a leakage-controlled forecasting pipeline, residual-based anomaly detection, model interpretation, and an interactive Streamlit dashboard.

The current version uses 180 days of synthetic hourly telemetry rather than measurements from a real facility. Its reported performance is therefore evidence that the software and analytical workflow operate correctly under controlled conditions; it is not evidence of production performance on an unseen building. The project should be presented publicly as an end-to-end prototype until a properly licensed real-world case study is added.

System Purpose and User Experience

The application provides three connected capabilities:

- Energy forecasting estimates the building's expected electricity use for each hour.
- Anomaly detection identifies hours whose actual consumption differs substantially from the forecast.
- Decision support presents forecasts, model drivers, operational interpretations, and what-if scenarios in a browser-based dashboard.

A building manager can inspect actual and predicted demand, select a flagged hour, review the size and direction of the error, and receive a cautious prompt such as reviewing after-hours schedules or HVAC operation. The application does not diagnose equipment failure and does not claim that every unusual reading represents a fault.

End to End Workflow

Step	Component	Output
1	Hourly telemetry	Weather, occupancy, time, subsystem loads, and total energy
2	Validation	Parse timestamps, verify required fields, sort records, and reject invalid values
3	Feature engineering	Create cyclic time variables and strictly historical energy features
4	Chronological split	Train on 70 percent, calibrate on 10 percent, and test on the final 20 percent
5	Forecasting	Random Forest estimates expected hourly energy consumption
6	Anomaly scoring	Residual threshold is calibrated before untouched holdout evaluation
7	Interpretation	Permutation importance and rule-based operational explanations are displayed

Data Used by the Application

The demonstration dataset contains 4,320 hourly observations covering January 1 through June 29, 2026. It is synthetic, deterministic, and reproducible from source code. A fixed random seed ensures that another user can regenerate the same data.

Column	Meaning	Model role
timestamp	Date and hour of observation	Indirectly
outdoor_temp_c	Outdoor temperature in degrees Celsius	Yes
humidity_pct	Relative humidity percentage	Yes
occupancy	Estimated number of people in the building	Yes
hvac_kwh	Hourly HVAC energy	Visualization only
lighting_kwh	Hourly lighting energy	Visualization only
plug_load_kwh	Computers and other plug loads	Visualization only
critical_load_kwh	Continuously operating critical loads	Visualization only
total_energy_kwh	Measured total hourly energy	Prediction target
injected_anomaly	Known simulated abnormal period	Verification reference

The **subsystem energy columns** are intentionally excluded from the forecasting features. Because they nearly sum to total energy, including them would make the task artificially easy and introduce target leakage.

How the Synthetic Data Is Generated

- Temperature combines seasonal change, a daily cycle, and random weather variation.
- Humidity is inversely related to temperature and includes natural variation.
- Occupancy is higher on weekdays and during normal commercial operating hours.
- HVAC demand rises during cold or hot conditions and increases modestly with occupancy.
- Lighting and plug loads rise during occupied hours, while critical loads remain relatively stable.
- Selected intervals receive elevated HVAC or plug-load consumption to simulate abnormal operation.

Example Input Observations

Time	Temp	RH	Occ	HVAC	Light	Plug	Critical	Total
Jan 01 00:00	-20.16	73.82	7	25.04	2.03	2.52	3.45	33.22
Jan 01 01:00	-22.79	80.70	12	26.91	0.99	3.05	3.62	34.20
Jan 01 07:00	-19.09	80.15	21	25.88	6.42	3.43	3.33	38.87

Prediction Target and Feature Engineering

The prediction target is `total_energy_kwh`, the total electrical energy consumed during an hour. The model combines current environmental and operational conditions with time indicators and prior energy consumption.

Predicted energy at time $t = f(\text{weather occupancy time and historical energy})$

Feature group	Variables	Purpose
Environmental	outdoor_temp_c, humidity_pct, occupancy	Current operating conditions
Cyclic time	hour sine and cosine	Preserves the closeness of hour 23 and hour 0
Calendar	weekday sine and cosine, month sine and cosine, weekend	Represents weekly and seasonal patterns
Historical	lag 1 hour, lag 24 hours, prior 24-hour mean	Represents recent and daily demand behaviour

Every lag and rolling feature is shifted before calculation. The value being predicted is never included in its own input features.

Machine Learning Technique

Random Forest Regression

The forecasting model is a Random Forest regressor containing 240 decision trees. Each tree learns different nonlinear rules from subsets of the observations and features. The forest prediction is the average of the individual tree predictions.

Forest prediction = sum of individual tree predictions divided by number of trees

Random Forest was selected because it:

- captures nonlinear relationships and interactions among weather, occupancy, time, and prior demand;
- performs well on tabular data without feature scaling;
- trains quickly enough for an interactive portfolio project;
- is more stable than a single decision tree; and
- can be interpreted using permutation importance.

The current system does not use a neural network, an auto-encoder, Isolation Forest, SHAP, or a language model. These techniques should not be claimed in a résumé or LinkedIn description of the current version.

Training and Holdout Evaluation

Randomly splitting time-series observations could allow future patterns to influence training. SmartBuild AI instead uses the first 70 percent for training, the next 10 percent for anomaly-threshold calibration, and the final 20 percent as an untouched chronological holdout.

Partition	Observations	Description
Original dataset	4,320	January 1 to June 29, 2026
Usable after lag creation	4,296	First 24 hours removed
Training portion	3,007	Earlier 70 percent
Calibration portion	429	Next 10 percent
Holdout portion	860	May 25 to June 29, 2026

Actual and Predicted Output

Actual output is the observed total_energy_kwh value. Predicted output is the Random Forest estimate. The residual is actual minus predicted. A positive residual means consumption exceeded the forecast; a negative residual means the forecast exceeded consumption.

Residual = actual energy minus predicted energy

Time	Actual kWh	Predicted kWh	Residual kWh	Flagged
May 25 04:00	14.74	14.79	-0.05	No
May 25 05:00	15.26	16.61	-1.34	No
Jun 29 22:00	15.21	14.26	+0.95	No
Jun 29 23:00	9.14	13.00	-3.86	No

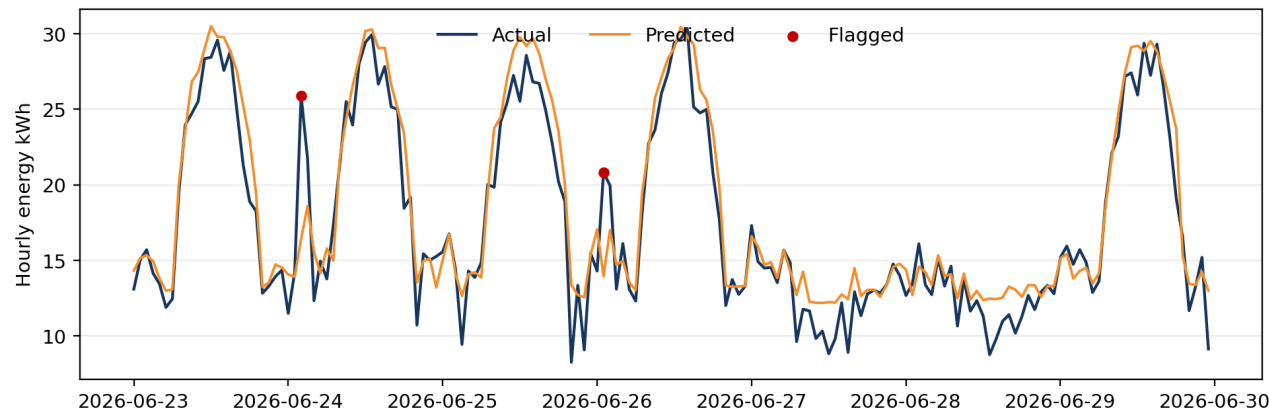


Figure 1 Actual and predicted hourly energy during the final seven holdout days

Residual Based Anomaly Detection

The model converts forecast error into an anomaly score. For the labelled synthetic demonstration, the threshold is selected on the separate calibration period to maximize F1 and is never tuned on the final holdout. The resulting threshold is 5.776 kWh. Unlabelled uploads use the 97.5th percentile of calibration residual magnitude.

$$\text{Anomaly score} = \text{absolute residual divided by } 5.776 \text{ kWh}$$

A score greater than 1 is flagged. Scores below 1 remain within the selected residual range.

Time	Actual	Predicted	Residual	Score	Injected
Jun 24 02:00	25.91	16.43	+9.48	1.64	Yes
May 27 07:00	26.00	18.54	+7.46	1.29	Yes
Jun 09 18:00	17.36	24.23	-6.87	1.19	No
Jun 26 01:00	20.81	13.96	+6.84	1.18	Yes
Jun 22 19:00	15.54	21.87	-6.33	1.10	No

A flagged residual is a screening signal rather than a fault diagnosis. High demand can arise from legitimate schedule changes, unusual occupancy, weather, sensor error, or equipment behaviour.

Performance Metrics

Metric	Result	Interpretation
Mean Absolute Error	1.397 kWh	Average absolute hourly prediction error
Root Mean Squared Error	1.836 kWh	Penalizes larger errors more strongly
R squared	0.913	Share of holdout variation explained by the model
Baseline MAE	11.160 kWh	Error from always predicting the training mean
MAE improvement	87.5 percent	Reduction in MAE relative to the baseline
Anomaly precision	71.4%	Share of flags matching injected events
Anomaly recall	29.4%	Share of injected events detected
Anomaly F1	41.7%	Balance of precision and recall

Mean Absolute Error

MAE = average of the absolute prediction errors

The model differs from actual hourly consumption by approximately 1.40 kWh on average.

Root Mean Squared Error

RMSE = square root of the average squared prediction errors

RMSE is higher than MAE because it places greater weight on occasional large errors.

Coefficient of Determination

An R squared value of 0.913 means that the model explains approximately 91.3 percent of the variation in the synthetic holdout data. It does not mean that the model is 91.3 percent accurate.

Anomaly Precision Recall and F1

On the untouched synthetic holdout, the detector achieved 71.4% precision, 29.4% recall, and 41.7% F1. It identified 5 of 17 injected anomalous hours and produced 2 false alerts. The strong precision and lower recall describe a conservative screen that misses some injected events.

Model Interpretation

SmartBuild AI currently uses permutation importance. Each feature is shuffled while the remaining inputs are left unchanged. If prediction error increases, the model depends on that feature.

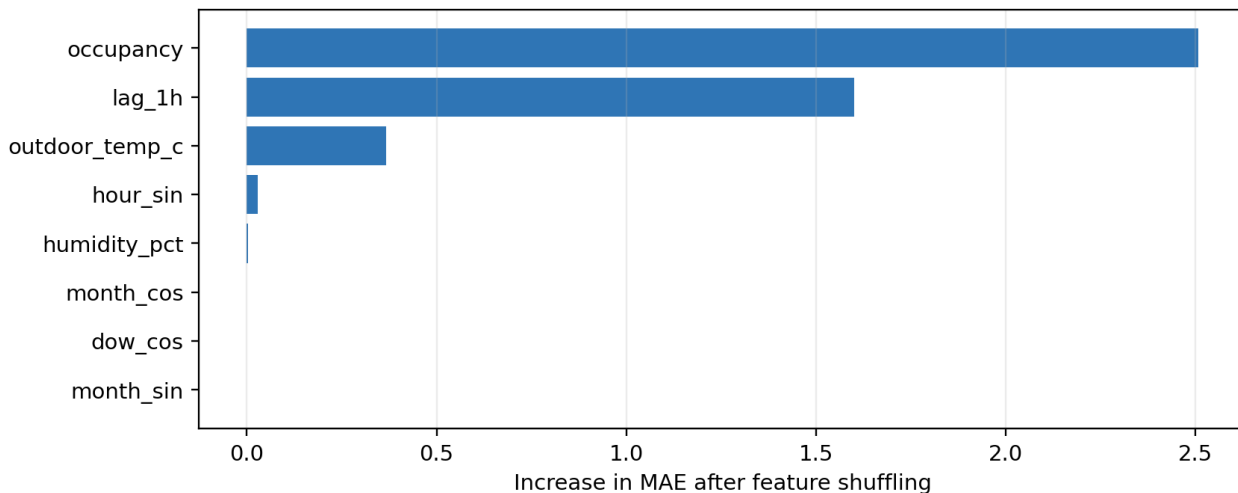


Figure 2 Leading model features measured by permutation importance

Rank	Feature	Importance
1	occupancy	2.507
2	lag_1h	1.601
3	outdoor_temp_c	0.367
4	hour_sin	0.029
5	humidity_pct	0.004
6	month_cos	-0.000
7	dow_cos	-0.001
8	month_sin	-0.001

Previous-hour consumption, occupancy, and outdoor temperature are the most influential model inputs. Importance describes model dependence and does not establish physical causation.

Operational Explanations

After an observation is flagged, a rule-based explanation examines residual direction, time of day, temperature, and occupancy. These explanations are deterministic decision-support messages rather than language-model outputs.

Observed pattern	Suggested review
High after-hours demand	Review HVAC, lighting, and plug-load schedules
High demand in extreme temperature	Inspect HVAC efficiency and setpoints
High demand with low occupancy	Check equipment left operating or schedule overrides
Consumption below expectation	Verify schedules before treating the reading as a fault

Software Components

Component	Responsibility
app.py	Interactive Streamlit dashboard
data.py	Synthetic data generation validation and cleaning
features.py	Cyclic time and historical feature engineering
modeling.py	Model training evaluation importance and persistence
anomalies.py	Anomaly scoring and operational explanations
generate_demo_data.py	Recreates the demonstration input data
train_model.py	Trains and evaluates the Random Forest
export_holdout_predictions.py	Exports inspectable out-of-sample predictions
tests	Automated checks for data features and modeling

Current Limitations

- The dataset is synthetic and does not establish performance on a real facility.
- Residual anomalies may reflect legitimate operational changes rather than equipment faults.
- The what-if simulator holds historical lag features constant and is not a physical building simulation.
- Anomaly metrics are reported only for synthetic injected labels and do not establish performance on real faults.
- Production use would require sensor-quality monitoring, drift detection, retraining controls, access control, and validation by facility professionals.

Recommended Next Improvements

Priority	Improvement	Value
1	Add a licensed real-world building-energy dataset	Provides an external case study
2	Add time-series cross-validation	Tests stability across multiple periods
3	Improve recall without sacrificing alert quality	Detects more injected abnormal periods
4	Add SHAP only if implemented and validated	Provides local prediction explanations
5	Deploy the dashboard publicly	Makes the live application accessible to recruiters